

Instructor Performance and Course Quality Evaluation on EduPro

1. Introduction

Online education platforms depend heavily on instructor quality and course design. High-quality teaching improves learner satisfaction, course ratings, repeat enrollments, and platform credibility. EduPro aims to establish a data-driven framework to evaluate instructor effectiveness and course quality.

This project analyzes instructor performance, teaching experience, expertise, and course ratings to identify factors that contribute to successful learning outcomes.

2. Objectives

The objectives of this project are:

- Analyze instructor rating distribution.
- Study the relationship between teaching experience and performance.
- Evaluate course quality across categories and levels.
- Identify top-performing instructors and expertise areas.
- Examine the impact of instructor ratings on course success.
- Provide recommendations for improving teaching effectiveness.

3. Dataset Description

The project uses three datasets:

Teachers Sheet

- TeacherID
- TeacherName
- Age
- Gender
- Expertise
- YearsOfExperience
- TeacherRating

Courses Sheet

- CourseID
- CourseName
- CourseCategory
- CourseLevel
- CourseRating

Transactions Sheet

- TransactionID
- CourseID
- TeacherID

4. Methodology

Data Integration

The datasets were merged using:

- TeacherID
- CourseID

This enabled mapping between instructors and courses.

Data Analysis

The following analyses were performed:

1. Instructor Profile Analysis
2. Experience vs Rating Analysis
3. Course Quality Evaluation
4. Expertise-Based Performance Analysis
5. Enrollment and Course Success Analysis

5. Exploratory Data Analysis

Instructor Rating Distribution

Teacher ratings were analyzed to identify:

- High-performing instructors
- Average-performing instructors
- Low-performing instructors

Most instructors maintained ratings above average, indicating overall teaching quality across the platform.

Experience vs Teacher Rating

Scatter plot analysis was used to evaluate the relationship between:

- Years of Experience
- Teacher Rating

Results indicate that experienced instructors generally receive higher ratings, although the relationship is not perfectly linear.

Course Category Performance

Course ratings were analyzed by category.

Findings indicate that some categories consistently achieve higher ratings due to stronger instructional quality and content structure.

Expertise Performance

Instructor expertise areas were compared against average ratings.

Certain expertise domains demonstrated consistently stronger performance and learner satisfaction.

6. Key Performance Indicators (KPIs)

Average Teacher Rating

Measures overall teaching effectiveness across the platform.

Average Course Rating

Measures overall course quality and learner satisfaction.

Rating Consistency Index

Evaluates instructor reliability and rating stability.

Experience Impact Score

Measures the influence of teaching experience on ratings.

Enrollment Influence Ratio

Evaluates whether highly rated instructors attract more enrollments.

7. Findings and Insights

Insight 1

Higher-rated instructors tend to be associated with better course ratings.

Insight 2

Teaching experience contributes positively to instructor performance.

Insight 3

Some expertise areas consistently outperform others in learner satisfaction.

Insight 4

Course quality varies across categories, suggesting the need for category-specific quality improvement strategies.

Insight 5

Top instructors contribute significantly to overall platform credibility and learner engagement.

8. Recommendations

1. Develop instructor training programs for low-rated instructors.
2. Encourage mentorship between experienced and new instructors.
3. Implement continuous performance monitoring.
4. Create quality standards across all course categories.
5. Reward top-performing instructors through recognition programs.
6. Use learner feedback to improve course design and teaching effectiveness.

9. Conclusion

This project successfully establishes a data-driven framework for evaluating instructor performance and course quality on EduPro. The analysis demonstrates that instructor expertise, teaching experience, and rating consistency significantly influence course success and learner satisfaction. By leveraging these insights, EduPro can improve educational outcomes, enhance platform credibility, and support continuous quality improvement.